NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.4c Thunderstorms: AI forecasts driving WRF

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 14:38 — 2026-10-24 21:52

Stage conclusion

Project report

Thunderstorm: AI forecasts driving WRF (K)

Objective

Use one WRF setup (4 km single domain, no cumulus parameterization) driven at the initial and boundary conditions by either Pangu forecasts or GFS forecasts. Start forecasts 24/48/72 h ahead for thunderstorm events that have already been simulated and have observations. Compare storm location and timing, radar objects and station gusts to judge whether Pangu driving is non-inferior to GFS driving. This serves as the accuracy benchmark for node L.

Current status

10-24 check: the R-group control runs (13-level GFS driving; option ① approved by the PI on 10-21) have all finished and been finalized. No jobs are running. - **Conclusion v2**: 6/6 valid reviews in the six-reviewer review; finalized under reading A. - For the single Akron event under this configuration, 13-level Pangu driving did not reach "non-inferior to GFS driving with the same number of levels" at any of the 24/48/72 h lead times (3/3). - The level-count effect (including the Q2 re-extraction) could be judged only at 24 h, and only at the margin: the position error exceeds the 50 km tolerance by 2.8 km. At 48 and 72 h the timing cannot be judged, so the two effects cannot be separated. - Do not write "AI is worse than GFS". - **Stop category**: remains On hold (awaiting PI). The remaining options are left to the PI: ② extend to ≥3 events; ③ find an AI model with full-level output; ④ stop further work.

Decisions made

- 2026-10-21/22 time-step harmonization: because R48 crashed at dt20, all 48 h runs moved to dt15 (G48/P48 rerun); because R72 crashed at dt15, all 72 h runs moved to dt12 (G72/P72 rerun). The original time steps are kept as supplementary columns; disagreeing verdicts are recorded as "time-step sensitive, no verdict". If R72 crashes again at dt12, 72 h is marked "cannot be separated".
- 2026-10-21 the PI approved option ① (R group, 13-level GFS control); experiment card written, with criteria locked at 10-21 09:45.
- 2026-10-17 Closed (negative result) ("worth continuing" not met): 0/3 non-inferior at 24/48/72 h, and both groups had 24 h position errors >150 km. Do not write "AI driving is worse than GFS" (model and level count were not decoupled; R24 had not been run). Restart conditions are in the decision record.
- 2026-10-15 protocol clarification (filling a design gap; criteria unchanged): the formal minimum area for ≥40 dBZ objects is ≥64 km^2, with no threshold and ≥16 km^2 reported alongside; if the three thresholds disagree, that lead time is marked sensitive and not counted toward "worth continuing". The timing search window now starts at 00Z (the position window stays 03–09Z); if an object already exists in the first 00Z frame, the timing metric is left-censored and not judged. Revised in v2 after review: a vote counts only when the 16 and 64 km^2 thresholds agree, otherwise neither a positive nor a negative vote counts; the area threshold applies to all object judgments; "within R km" uses the nearest grid cell; left-censored values enter the original formula as intervals, and no positive vote is counted if the result cannot be judged.
- 2026-10-14 design v2: the GFS group uses full vertical levels (Deng 2025 abstract: GFS truncated to 13 levels performs worst); 13-level GFS becomes the optional run R24.
- 2026-10-14 experimental design v1: the main Pangu runs start from GFS analyses (same initial state as the GFS group); ERA5-initialized runs are only the optional E24. Both groups use the same 13 levels, 6 h boundary interval and land surface; a 6 h anchor pair and a gate are added.
- 2026-10-14 node opened. First event: Akron 2020-06-09; Wister 2018-08-10 serves only as an initiation control. WRF runs on CRC (no SU charged) or on a reduced domain.

Attempts and results

Date	What was done	One-line result
2026-10-22	Formal 48 h pair at dt15 plus supplementary G48/P48 at dt20; judged by the lead with a scoring script	(a) R48 vs G48: position 79.8 ≤ 379.6 ✓; timing for R is left-censored [190,∞) vs 230, so it cannot be judged, no verdict; (b) P48 vs R48: position 206.9 > 129.8, not non-inferior (consistent across formal/supplementary runs and both thresholds); 48 h effects cannot be separated
2026-10-22	R24 integration on CRC (dt20) plus three tables (G24 regression identical cell by cell); judged by the lead per item ⑤ of the experiment card	(a) R24 vs G24: position 236.2 > 233.4 (over by 2.8 km) *X*, timing 0 ≤ 110 ✓, not non-inferior (both thresholds agree); (b) P24 vs R24: at the 64 km² threshold position 307.9 > 286.2 and timing 170 > 60 *X*; at the 16 km² threshold timing ≥160 > 90 *X*; not non-inferior. R24 maximum 10 m wind within 100 km is 20.5 m/s (0.55 of the observed 37.04 m/s gust)
2026-10-21	wx generated the 13-level GFS driving fields (CRC)	30 files with SHA manifest; all checked by the lead at 14:27; read-back PASS
2026-10-17	The WRF line delivered the 72 h pair (formal G72/P72 at dt15 on CRC; dt20 as a supplementary column) and added 150 km existence checks; the lead tallied votes	All four 72 h judgments agree: P72 position 307.7/299.7 km vs G72 104 km; at 64 km² G has an object and P has none, so not non-inferior. All three lead times count as "clearly worse" (48 h on timing alone). Per the conclusion rule, drafted "shelve (AI driving worse than GFS, single-event, qualitative)"; sent to 6 reviewers; status unchanged until the ruling
2026-10-15	The WRF line delivered the formal v2 table for G48/P48	At ≥64 km²: G48 first appears at 00:30Z (−160 min), 06:10Z centroid error 316.8 km, object 80 km², maximum wind 17.1; P48 has no first appearance within 50 km in the window (right-censored), error 273.3 km, object 287 km², maximum wind 17.0. At ≥16 km²: G48 −140 min / 275.2 km; P48 no first appearance / 174.7 km. CSV checked by the lead. Asked for a 150 km existence column for each pair; judgment waits for the 72 h pair (about 10-18)

Date	What was done	One-line result
2026-10-15	The WRF line accepted the pairing plan	Formal 72 h pair = G72 dt15 vs P72 dt15 (P72 running); G72 dt15 gets a new CRC run root and is submitted after the WRF line's review, expected to finish late on 10-17; G72 dt20 kept as a supplementary column; the 48 h pair to be delivered on the evening of 10-15
2026-10-15	The WRF line delivered the formal v2 table for G24/P24	At ≥64 km^2: first appearance observed 03:10Z, G24 04:00Z (+50 min), P24 00:20Z (−170 min); 06:10Z centroid error G24 183.4 / P24 307.9 km; maximum 10 m wind 22.2 / 21.5 m/s. At ≥16 km^2: P24 is left-censored (object in the first 00Z frame) and a different 32 km^2 object is selected, position 233.1 km. CSV checked by the lead. Judgment waits for all three pairs; for the 72 h pair, P72 moved to dt=15 s because of CFL problems, and G72 was asked to rerun at the same dt
2026-10-15	The WRF line delivered the formal v2 table for G06/P06	At ≥64 km^2: first appearance observed 03:10Z, G06 04:40Z (+90 min), P06 00:50Z (−140 min, within spin-up, anchor only); 06:10Z centroid error G06 53.1 / P06 66.6 km; object area observed 1380, G06 928, P06 1055 km^2; maximum 10 m wind within 100 km 24.8 / 26.7 m/s vs observed gust 37.0. G06 passes the gate under v2 (a ≥64 km^2 object within 150 km). At ≥16 km^2, P06 selects a different 48 km^2 object. The IEM N0Q cross-check shows overlapping positions but a different order of magnitude in area; used only as a check with stated limits. Accepted by the lead; protocol unchanged
2026-10-15	The WRF line adopted protocol v2	New script version (old version kept) judges hit50/150 km/gate after area filtering; G06 gate rechecked at ≥64 km^2. Formal tables for G06/P06 come first; paths reported for each other pair; the longest group finishes about the morning of 10-16; fallback follow-up at 10-16 16:41 still set

Date	What was done	One-line result
2026-10-15	The WRF line adopted two protocol clarifications	Formal table covers 3 area thresholds × 2 window start times. G06/P06 come first; the other 6 integrations finish by about the morning of 10-16. The observed ≥40 dBZ object at 06:10 is only one 2 km grid cell, suggesting an error in the three-radar composite; being cross-checked against IEM N0Q (if confirmed, the observations may have no object under the formal protocol, which must be reported as is and judged by the lead)
2026-10-15	ideas generated driving fields v2.2 (CRC job, CO-, 0 SU)	All 66 files pass the SHA check (rechecked by the lead); only Q2 changed, everything else is byte-identical to v2.1. Within 300 km of KAKO, new minus old is −0.37 to −0.91 g/kg, exactly the opposite sign of the WRF line's report, i.e. equal to the reference algorithm
2026-10-14	WRF line G06/P06 preprocessing trial	metgrid passed (G 32 levels / P 14 levels); ocean SNOW=0, SMOIS=1, SST=TSK confirmed; real.exe produced NaN in vertical interpolation in the boundary zone at 06-09 12Z for both groups (judged by the WRF line to be a problem in their own pipeline); Q2 sensitivity: v2.1 is too moist by +0.38 to +0.91 g/kg, maximum absolute difference 1.6 to 9.3 g/kg, all above 1 g/kg, so moved to v2.2
2026-10-14	ideas generated 8 sets of driving fields (v2.1, 0.25° GFS)	66 WPS files, 1.6 GB; read-back and SHA checks pass (lead spot-checked 4, OK); G 31 levels / P 13 levels; land surface byte-identical between groups; ocean soil and snow filled from the nearest cell and negative Pangu q clipped to 0 (both labeled assumptions)
2026-10-14	ideas pre-check of driving fields	Multi-lead 2020 GFS 0.5° forecasts are no longer publicly archived (NCEI 404, keeps about 2 years; AWS keeps only 30 days); native 0.25° is downloadable from NCAR GDEX d084001; Colab not used, 0 CU

Date	What was done	One-line result
2026-10-14	Grok quick search for precedents	No precedent for AI-driven convection-permitting models for mid-latitude downbursts or thunderstorm winds. Deng 2025 abstract checked: 33-level GFS ≈ 13-level Pangu, and GFS truncated to 13 levels is worst. Brewer 2025 abstract checked: GraphCast-driven WRF is comparable to GFS/ECMWF-driven WRF (downslope winds)

Parameter sources

Parameter	Value	Source or "assumption"	Places checked
WRF d01 setup	4 km, 748×600×80, dt 20 s, cu0, mp8, PBL2/sf2, lw4	Authors' namelist	WRF lead reply
Compute cost	About 1 SU per model hour (not benchmarked)	WRF lead, scaled from an earlier experiment	Short test pending
Projection	Lambert, 38N/100W, truelat 38/38	WRF line; authors' wrfinput metadata	—
grid_fdda	0	Authors' namelist.input, line 113	—
GFS source for this experiment	Native 0.25°, NCAR GDEX d084001	ideas pre-check	NCEI, AWS and d335000 have no 2020 0.5° forecasts
RH→q (G group), Pangu negative-q clipping, nearest-cell fill of ocean soil/snow	See preprocessing notes	**Assumption** (implementation)	To be tested by metgrid/real
2 m specific humidity	v2.2: q from the lowest complete pressure level with GHT > SOILHGT (v2.1 ln p interpolation withdrawn)	**Assumption** (design v2.2; based on the WRF line's Q2 sensitivity test; no literature basis)	Brewer repository: its GFS input gives no 2 m RH
Authors' driving data	GFS 0.5°, 00Z initialization, 3-hourly	WRF line	—
Observed object source	NEXRAD Level-II column-maximum reflectivity composite + KAKO METAR/special reports; gust anchored at 37.04 m/s	WRF line, 10-15 (1-minute ASOS missing that day; NCEP archive has no MRMS for that day)	Criteria thresholds unchanged

Parameter	Value	Source or "assumption"	Places checked
Observed peak time	About 06:10 UTC (02:10 EDT at KAKO)	Card GZYPXN4S, PDF pp.9–13 (cited via the WRF line's final report)	To be checked by this node
GFS benchmark levels	Full levels (not truncated to 13)	Deng 2025 AOSL 10.1016/j.aosl.2025.100589 abstract (original abstract read via OpenAlex)	Full text not read (ScienceDirect 403)
Pangu initial state	GFS analysis at the same time	Not stated in the Deng 2025 abstract; Grok says its SI specifies GFS analyses (**not verified**)	ideas to check the original
Surface/soil fill, 2 m humidity, PSFC	Land surface from GFS f000 at initialization; 2 m humidity derived the same way for both groups; PSFC not supplied	**Assumption** plus WRF lead advice; the Brewer 2025 repository rfovell/GraphCast-WRF (Apache-2.0, last push 2024-12-19) has a fill method still to be checked	Delegated
WRF version	4.0.3	WRF lead (the authors' results are version-sensitive)	—

Open items and next steps

- ideas delivers the driving fields → check the field list and parity → formally announce the path.
- The Grok quick search has returned (see attempts and results). Deep Research step 2 **not done**: the 30-day quota (30/30) is used up; to be sent when the quota resets or the PI adds quota.
- The WRF lead accepted v2: the conclusion must state that the main comparison mixes model and level-count differences; with spare capacity, run R24 first, then E24; one metgrid/real trial each for G06 and P06.
- Driving fields v2.2 delivered (10-15 05:00, reported to the parent node and forwarded by the root lead to the WRF lead).
- The WRF line started runs after rechecking v2.2 at 10-15 04:58: the boundary NaN in real is avoided with all_ic_times=true (tested to have bit-for-bit no effect on boundary forcing). G06/P06 run first (about half a day) and the G06 gate is judged per section 5. After the gate, the 6 integrations of the 24/48/72 pairs run in parallel (the longest, 78 h group takes about 16–18 h wall clock, plus queue time). 3D fields are kept only for 03–09Z, target <15 GB. Numerical tables are stored and returned to this node, which judges them under the frozen criteria and reports to the parent node. Fallback follow-up at 10-16 16:41.
- Waiting on: design objections, delivery location, schedule.
- WRF line short test → 06 pair passes the gate → 24/48/72 pairs → comparison table → conclusion (before it goes to the PI).

Boundaries

- Testbed rules: no per-experiment limit on Colab (the parent project corrected this on 10-14 per the PI's words "use Colab heavily for now, no CU limit"; the earlier ≤20 CU cap is void). If one experiment is expected to exceed 100 CU, stop and report to the parent node lead first. Always run colab stop when done so sessions do not sit idle; dispatch only one executor per round. Supercomputer jobs ≤50 SU each (CRC does not charge SU). WRF jobs are submitted by the WRF line's agent and need a ComputeOps resource request, registration, and separate commands for pre-check and submission. Other lines' data are read-only.
- Storage and compute locations (tightened by the PI on 10-14): the workstation does not run computations or store compute data; it holds only scripts, small results under 200 MB each, figures and reports. If temporary storage over 1 GB is truly needed, report to the parent node lead first. Driving fields and WRF output stay on Colab or the clusters (currently CRC); new large data go preferably to TACC (the CRC group space is about 95% full), and CRC defaults to a single node. Every task brief must state where the computation runs and where results are stored.
- Hierarchy (set by the root lead on 10-14, 5-level cap): this is a level-2 node and **will not open level-3 sub-projects**. If subdivision is needed, use tasks within this node or ask the parent node lead (thunderstorms and convective scale) to open a sibling level-2 node.
- Only two hops remain downward: this node → agent → tool. Task briefs must not let tool agents (gpu-gemini, reviewers, litwiki and others) delegate further.
- The subtree list is not grouped by parent node; the project tree in the interface is authoritative.

Literature basis

- Akron mechanism: card 2DPBW6VX, PDF pp.6, 9–11; observations: Childs 2021, card GZYPXN4S Table 2, PDF p.16 (cited via the WRF lead, to be checked by this node).
- Xu 2024 ERL 10.1088/1748-9326/ad41f0; Liu 2024 JGR:MLC 10.1029/2024JH000207 (no card yet; not in the local library).
- Deng 2025 AOSL 10.1016/j.aosl.2025.100589 (abstract checked, no card yet); Brewer 2025 WAF 10.1175/WAF-D-24-0097.1 (abstract checked, no card yet). These 4 papers were delegated to litwiki to create cards and check the full texts (10-14 reply: PDFs not obtained; passed to Literature to fetch full texts, cards to follow). Not cited in conclusions until cards exist.

PI notes

- 10-14: fundamentally different in scale from typhoons (B3); kept as a separate project.

10-23 update: 72 h judgment and combined draft for the three lead times

- 72 h (formal dt12; WRF read-out pass_all; R72 did not crash at dt12): the level-count timing cannot be judged (none of the three runs has a first appearance in the window), so no verdict; the model comparison is not non-inferior (P72 290.8 km vs R72 100.4 km); 72 h effects cannot be separated (reading B: levels non-inferior).
- Combined draft for the three lead times: model comparison 3/3 not non-inferior; level count judged only at 24 h, at the margin. Sent to the six-reviewer panel (to rule on reading A vs B and the stop category).
- 10-23 six-reviewer ruling: reading A (6/6). Finalized: model 3/3 not non-inferior; level count (including the Q2 re-extraction) judged only at 24 h, at the margin; effects cannot be separated at 48/72 h. **Stop category: On hold (awaiting PI)** (option ① completed; options ② extend to ≥3 events, ③ AI model with full-level output, ④ stop further work, are left to the PI).

Key figures